EXPERIMENT - 12

AIM:

To implement echo client server using TCP/UDP sockets program.

CODE : (TCP)

```
import socket
import Threading
# --- server--.
def start_server():
    server =socket.socket(socket.AF_INET.
                            socket.sock_STREAM)
     server.bind (('127.0.0.1',5000)
      server.listen(1)
   Print ("[SERVER ]Listening on 127.0.0.1:
                                      5000")
     conn, addr = server.accept()
      Print (f"[SERVER]  Connected by {addr}
 while True:
     data = conn.recv(1024).decode()
    if not data or data. lower()=='quit':
           Print ("[Server] Connection closed
                                  by client.")
        break
    Print (f"[SERVER] Received :{data}")
       conn.send (f"server received : {data}".encode())
  conn.close()
  server.close()
```

```
#--CLIENT--
def start_client():
        client = socket.socket (socket.AF_INET,
                                  socket.SOCK
                                    _STREAM)

        client.connect (('127.0.0.1', 5000))
        Print ("[CLIENT] connected to server")

while True:
    msg = input ("Enter message (or 'quit' to exit):")
    client.send (msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv (1024).decode()
    Print (f"[SERVER RESPONSE ]{response}")
    client.close()

#MAIN
if __name__ == "__main__":
        # Run server in a background Thread
        server_thread = threading.Thread
                (target = start_server,
                      daemon = True)

        server_thread.start()
        # Run client in main Thread
        start_client()
```

output:

```
[SERVER] Listening on [127.0.0.1:5000 [CLIENT] connected
to server (127.0.0.1, 53455) Enter message
                               (or 'exit' to exit):
[SERVER] Connected by
Hello.
```

```
[SERVER] Received : Hello
[SERVER RESPONSE ] Server Recived : Hello
Enter message (or 'quit' to exit ): quit
[SERVER ] Connection closed by client
```

UDP:

```
import socket
import threading

# --SERVER--
def udp.server():
        s= socket.socket (socket.AF-INET,
                          socket.SOCK.DGRAM)
          s, bind ((127.0.0.1', 12000))
          Print ('[SERVER] Running on 127.0.0.1
                                           :12000)

   while True:
        data.addr = s.recvfrom(1024)
        msg = data.decode ()
        if msg.lower() = 'quit':
             Print (f"[SERVER] client {addr}
                          ended the connection.")
             break
        Print (f"[SERER] Recived from {addr}:
                                       {msg}")
        s.sendto (data.addr)
        s.close ()
        Print ( [SERVER] stopped.")
```

```
#---CLIENT---

def udp.Client()
    c= socket.Socket(Socket.AF_INET,
                     Socket.sock-DGRAM)

    server = ('127.0.0.1; 12000)
    while true:
        msg = input("Enter message ('or' quit' to
                                      stop ):")
        c.sendto (msg.encode(), server)
        if msg.lower = = 'quit'
            Print ("[CLIENT] Closing connection..")
            break
        data, _ = c.recvfrom(1024)
        Print ("Echo from server:", data.decode())
    c.Close()

#---- MAIN--

if __name__ == "__main":
    threading.Thread(target=udp_server,
                     daemon=True).start()

    udp_client()
```

OUTPUT:

```
Enter message (or quit) to stop): [SERVER] Running on
127.0.0.1:12000 Hi
[SERVER Recived from ('127.0.0.1', 49820): Hi
Echo from server: Hi
Enter message (or 'quit to exit): quit
[CLIENT] Stopped. [SERVER] stopped.
```

Result:
Thus the program to implement echo client server
using TCP/UDP socket was executed successfully